

Effects of a 1:1 inspiratory to expiratory ratio on respiratory mechanics and oxygenation during one-lung ventilation in the lateral decubitus position

S. H. KIM*, Y. S. CHOI†, J. G. LEE‡, I. H. PARK§, Y. J. OH**

Department of Anaesthesiology and Pain Medicine, Anaesthesia and Pain Research Institute, Yonsei University College of Medicine, Seoul, Korea

SUMMARY

Prolonged inspiratory to expiratory (I:E) ratio ventilation may have both positive and negative effects on respiratory mechanics and oxygenation during one-lung ventilation (OLV), but definitive information is currently lacking. We therefore compared the effects of volume-controlled ventilation with I:E ratios of 1:1 and 1:2 on respiratory mechanics and oxygenation during OLV. Fifty-six patients undergoing thoracoscopic lobectomy were randomly assigned volume-controlled ventilation with an I:E ratio of 1:1 (group 1:1, n=28) or 1:2 (group 1:2, n=28) during OLV. Arterial and central venous blood gas analyses and respiratory variables were recorded 15 minutes into two-lung ventilation, at 30 and 60 minutes during OLV, and 15 minutes after two-lung ventilation was re-initiated. Peak and plateau airway pressures in cmH₂O [standard deviation] during OLV were significantly lower in group 1:1 than in group 1:2 ($P < 0.01$) (19 [3] and 23 [4]; 16 [3] and 19 [5], respectively). The arterial to end-tidal carbon dioxide tension difference was significantly lower in group 1:1 than in group 1:2 ($P < 0.01$), (0.5 [0.3] and 1.1 [0.5]). There were no significant differences in P_aO_2 during OLV between the two groups (OLV₃₀, $P=0.856$; OLV₆₀, $P=0.473$). In summary, volume-controlled ventilation with an I:E ratio of 1:1 reduced peak and plateau airway pressures improved dynamic compliance and efficiency of alveolar ventilation, but it did not improve arterial oxygenation in a substantial manner. Furthermore, the associated increase in mean airway pressure might have reduced cardiac output, resulting in a lower central venous oxygen saturation.

Key Words: one-lung ventilation: inspiratory-expiratory ratio, oxygenation, respiratory mechanics

Oxygenation deterioration and alteration of respiratory mechanics are serious problems during one-lung ventilation (OLV) in the lateral decubitus position for thoracic surgery¹. High peak and plateau airway pressure during OLV is known as the strongest risk factor for acute lung injury and poor postoperative prognosis². Also, compression of the dependent lung results in atelectasis and alveolar collapse³, worsening of ventilation to perfusion (V/Q) mismatching, increased intrapulmonary shunt and consequent deterioration of oxygenation^{1,4-6}. Prolonged inspiratory to expiratory (I:E) ratio ventilation (e.g. inverse ratio

ventilation) is a concept that can be applied to patients with acute lung injury and acute respiratory distress syndrome to improve oxygenation, but the effects of this ventilatory strategy on respiratory mechanics and oxygenation have not been completely elucidated⁷⁻¹⁰. Prolonged I:E ratio ventilation, in which the lung is held at end-inspiratory volume for a longer part of the respiratory cycle, has been proposed to keep more alveoli in an inflated state and may help to prevent atelectasis and alveolar collapse by increasing the mean airway pressure, in addition to contributing to a decrease in peak airway pressure^{8,11,12}. Contrarily, this method has been reported to result in a decrease in cardiac output due to increasing mean airway pressure^{9,10}, which could possibly produce a negative effect on oxygenation during OLV⁵. Prolonged I:E ratio ventilation has a possibility of both positive and negative effects on respiratory mechanics and oxygenation during OLV, but definite information is currently lacking. The purpose of our study was to compare the effects of minimal prolonged 1:1 I:E ratio ventilation on respiratory mechanics and oxygenation

* MD, Clinical Assistant Professor.

† MD, PhD, Assistant Professor.

‡ MD, PhD, Assistant Professor, Department of Thoracic and Cardiovascular Surgery, Yonsei University College of Medicine.

§ MD, Resident.

** MD, PhD, Associate Professor.

Address for correspondence: Dr Y. J. Oh, Associate Professor, Department of Anaesthesiology and Pain Medicine, Anaesthesia and Pain Research Institute, Yonsei University College of Medicine, 250 Seongsanno, Seodaemun-gu, 120-752 Seoul, Korea. Email: yjoh@yuhs.ac

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with the conventional 1:2 I:E ratio ventilation during OLV in the lateral decubitus position in patients undergoing thoroscopic lobectomy.

METHODS

This study was approved by the Institutional Review Board of Severance Hospital, Yonsei University Health System (ref: 1-2011-0058) and registered at <http://clinicaltrials.gov> (ref: NCT01499212). After written informed consent was obtained from all participants, 56 patients aged 30–79 years with American Society of Anesthesiologists physical status I to II were scheduled for video-assisted thoracoscopic lobectomy requiring OLV under general anaesthesia were enrolled in this study. Patients with a history of coronary artery occlusive disease, chronic obstructive or restrictive pulmonary disease, cerebrovascular disease, renal insufficiency, heavy smoking or obesity (body mass index $>30 \text{ kg/m}^2$) were excluded. All patients underwent preoperative lung spirometry prior to surgery and those with a forced expiratory volume in one second, forced vital capacity and diffusion capacity of carbon monoxide less than 60% of the predicted value were also excluded. This randomised controlled trial took place at the operation centre of Severance Hospital in Seoul, Korea, from June to November 2011. Patients were randomly assigned to one of the two groups according to a computer-generated random numbers table. Patients received either an I:E ratio of 1:1 (group 1:1) or 1:2 (group 1:2) during OLV.

Standard monitoring devices were applied upon arrival at the operating room. Anaesthesia was induced with 1.5 mg/kg propofol and 1.0 $\mu\text{g/kg}$ remifentanyl. Tracheal intubation with a left-sided double-lumen tube (Broncho-Cath[®]; Mallinckrodt Medical Inc., Athlone, Ireland) was facilitated with 0.9 mg/kg rocuronium and the position of the double-lumen tube was confirmed with a fiberoptic bronchoscope both before and after turning the patient to the lateral decubitus position. After induction of anaesthesia, a 20-gauge radial artery catheter was installed and a 7-Fr central venous catheter (Arrow International, Reading, Pennsylvania, USA) was inserted via the right internal jugular vein. The central venous catheter length to be inserted was calculated using a height-based formula for its constant placement near the right atrium¹³. The placement of the tip of the central venous catheter was confirmed by portable chest X-ray. Anaesthesia was maintained with 1.0–2.0% sevoflurane, remifentanyl 0.1–0.3 $\mu\text{g/kg/min}$ and rocuronium 5.0–10.0 $\mu\text{g/kg/min}$.

The lungs of all patients were initially ventilated with a constant-flow volume-controlled ventilation

(VCV) mode (Zeus ventilator, Dräger Medical, Lübeck, Germany) with a tidal volume of 8 ml/kg, an I:E ratio of 1:2, and a respiratory rate of 8–12 breaths per minute in 100% oxygen with no positive end-expiratory pressure (PEEP) and no inspiratory pause. The respiratory rate was adjusted to maintain an end-tidal CO_2 tension ($\text{PE}'\text{CO}_2$) of 4.0–4.6 kPa. All patients were turned to the lateral decubitus position and two-lung ventilation (TLV) was conducted for 15 minutes before OLV. The ventilator settings were the same for TLV and OLV, except for the I:E ratio of the two groups during OLV. OLV was initiated with an I:E ratio of 1:1 (group 1:1) or 1:2 (group 1:2). All measurements were performed with the patient in the lateral decubitus position. Haemodynamic variables, respiratory variables, and arterial and central venous blood gas analyses were recorded at four time points: 15 minutes after placing the patient in the lateral decubitus position under TLV before OLV ($\text{TLV}_{\text{baseline}}$), 30 minutes after initiation of OLV (OLV_{30}), 60 minutes after initiation of OLV (OLV_{60}) and 15 minutes after TLV was resumed (TLV_{end}).

Haemodynamic measurements included heart rate, mean arterial pressure and central venous pressure. Respiratory variables included peak airway pressure (P_{peak}), plateau airway pressure (P_{plat}), mean airway pressure (P_{mean}) and dynamic compliance (C_{dyn}). $\text{PE}'\text{CO}_2$ was measured by capnography implemented in the ventilator. The oxygen content (CxO_2) in arterial and central venous blood was calculated using the following equation: $\text{CxO}_2 = 1.36 \text{ Hb SxO}_2 + 0.0031 \text{ PxO}_2$, in which Hb = haemoglobin concentration (g/dl) and SxO_2 = oxygen saturation. The alveolar-arterial O_2 gradient (A-aO_2) was calculated as the difference between alveolar oxygen tension (PAO_2) and arterial oxygen tension (PaO_2). Qs/Qt was determined using the following formula: $\text{Qs/Qt} = (\text{CcO}_2 - \text{CaO}_2) / (\text{CcO}_2 - \text{CvO}_2)$, where CcO_2 = calculated capillary O_2 content, assuming that the pulmonary capillary O_2 partial pressure is equal to PAO_2 and the central venous oxygen saturation (ScvO_2) is equal to the mixed venous oxygen saturation (SvO_2). Physiological dead space (Vd/Vt) was calculated according to the Hardman and Aitkenhead equation: $\text{Vd/Vt} = 1.14 \times (\text{PaCO}_2 - \text{PE}'\text{CO}_2) / \text{PaCO}_2 - 0.005$ ¹⁴. Arterial and central venous blood samples were analysed using an automated blood gas analyser (Stat Profile[®] CCX, Nova Biomedical, MA, USA). This study was planned to terminate if mean arterial pressure decreased more than 20% relative to the post-induction value requiring administration of a vasoactive drug, if SpO_2 as measured by pulse oximetry declined to less than 90%, or if PaO_2 decreased to less than 8.0 kPa during OLV.

Statistical analyses were performed with the Statistical Package for the Social Sciences 18.0 (SPSS Inc, Chicago, IL, USA). All data are expressed as mean $\pm$ standard deviation or the number of patients. Because no previous study has evaluated the effect of ventilation with different I:E ratios during OLV in the lateral decubitus position, the sample size calculation was based on a previous study that compared the effect of ventilation with different I:E ratios in the supine position¹⁰; 80% power to detect a 3 kPa difference in P_aO_2 between the groups with a standard deviation of 10% and an alpha level of 0.05 using an independent t-test required 26 patients in each group. Demographic data comparison between the groups was performed using the chi-square test, Fisher's exact test or an independent t-test as appropriate. Data comparison between the two groups at each time point and changes between time points within a group were compared using two-way repeated measure analysis of variance. Adjustment for multiple comparisons was made using Bonferroni's correction. A P value <0.05 was considered statistically significant.

RESULTS

Of the 62 patients initially assessed, 59 were enrolled in the study. Of the enrolled patients, three were excluded from the study due to changes in the surgical plan during surgery; therefore 56 patients completed all the assessments for our study (28 in group 1:2 and 28 in group 1:1). Physical characteristics, results of the preoperative pulmonary function studies and surgical data for these patients are listed in Table 1. There were no statistically significant differences between the two groups. None of the patients developed dangerous hypoxaemia or hypotension during OLV and the study was successfully completed in all patients.

The arterial and venous blood gas data at each time point and changes in P_aCO_2 , $Pa-E'CO_2$ difference, $ScvO_2$ and Qs/Qt are shown in Table 2 and Figure 1. P_aO_2 , $A-aO_2$ gradient, $PcvO_2$ and $ScvO_2$ decreased significantly in both groups during OLV (OLV₃₀, OLV₆₀), compared with TLV_{baseline} ($P < 0.01$). There were no significant differences in P_aO_2 between the two groups during OLV (OLV₃₀, $P=0.856$; OLV₆₀, $P=0.473$). P_aCO_2 , $PcvO_2$ and $Pa-E'CO_2$ difference and estimated Vd/Vt decreased significantly in group 1:1 compared with group 1:2 during OLV (OLV₃₀ and OLV₆₀, $P < 0.01$). PAO_2 increased significantly in group 1:1 compared with group 1:2 during OLV (OLV₃₀ and OLV₆₀, $P < 0.01$). $ScvO_2$ and Qs/Qt decreased significantly in group 1:1 compared with group 1:2 during OLV (OLV₃₀, $P=0.03$; OLV₆₀, $P=0.01$ and

TABLE 1

Patient characteristics, results of preoperative pulmonary function studies and surgical data

	Group 1:2 (n=28)	Group 1:1 (n=28)	<i>P</i> value
Age, y	61.7 (30–74)	60.3 (38–78)	0.615
Male/female	14/14	20/8	0.171
Height, cm	161.5 (9.0)	164.1 (6.9)	0.228
Weight, kg	61.5 (9.5)	63.0 (9.6)	0.567
Body mass index, kg/m ²	23.6 (2.6)	23.5 (3.0)	0.907
FEV1, %	98.0 (22.5)	99.4 (15.8)	0.795
FVC, %	95.0 (12.8)	93.6 (13.1)	0.704
FEV1/FVC, %	76.6 (7.9)	76.4 (7.6)	0.943
DLCO, %	101.9 (16.7)	106.5 (22.4)	0.443
Haemoglobin, g/dl	11.8 (1.2)	12.1 (1.3)	0.302
Right/left lobectomy	16/12	20/8	0.403
OLV time, min	144.6 (106.9)	128.7 (54.4)	0.483

Values are mean (range), mean (standard deviation) or number. FEV1=forced expiratory volume in one second, FVC=forced vital capacity, DLCO=diffusion capacity of lung for carbon monoxide, OLV=one-lung ventilation.

OLV₃₀, $P=0.03$; OLV₆₀, $P=0.02$, respectively).

Respiratory mechanics and haemodynamic data at each time point and changes in airway pressures and dynamic compliance are shown in Table 3 and Figure 2. Compared with initial TLV (TLV_{baseline}), the initiation of OLV in both groups was associated with a significant increase in P_{peak} ($P < 0.01$), P_{plat} ($P < 0.01$), P_{mean} ($P < 0.01$) and a significant decrease in C_{dyn} ($P < 0.01$). During OLV (OLV₃₀, OLV₆₀), P_{peak} and P_{plat} were significantly lower in group 1:1 than in group 1:2 ($P < 0.01$), but P_{mean} was significantly higher in group 1:1 than in group 1:2 ($P < 0.01$). C_{dyn} increased significantly in group 1:1 compared with group 1:2 during OLV ($P < 0.01$). There were no significant differences between the two groups in terms of respiratory rate and haemodynamic variables measured throughout the study period.

DISCUSSION

In the present study, VCV with an I:E ratio of 1:1 showed a favourable performance in terms of respiratory mechanics for OLV; this simple ventilatory strategy effectively reduced P_{peak} and P_{plat} , increased dynamic compliance and improved the efficiency of alveolar ventilation compared with conventional VCV with an I:E ratio of 1:2 during OLV. However, the overall systemic oxygenation did not improve with VCV at an I:E ratio of 1:1 due to a decrease in both intrapulmonary shunt and $ScvO_2$.

TABLE 2
Arterial and venous blood gas analysis data

	TLV _{baseline}		OLV ₃₀		OLV ₆₀		TLV _{end}	
	Group 1:2	Group 1:1	Group 1:2	Group 1:1	Group 1:2	Group 1:1	Group 1:2	Group 1:1
pH	7.42 (0.04)	7.43 (0.02)	7.41 (0.04)*	7.44 (0.03)†	7.40 (0.04)*	7.44 (0.04)†	7.40 (0.04)*	7.41 (0.04)
P _a O ₂ (kPa)	52.9 (10.3)	53.6 (9.6)	27.9 (9.1)*	27.5 (9.3)*	29.1 (10.7)*	31.1 (10.2)*	50.0 (10.8)	50.0 (12.3)
PcvO ₂ (kPa)	7.1 (1.2)	6.6 (1.3)	6.1 (1.1)*	5.2 (0.9)*†	6.0 (1.2)*	5.2 (0.7)*†	6.9 (1.0)	6.6 (1.4)
P _a CO ₂ (kPa)	4.8 (0.7)	4.6 (0.5)	5.0 (0.8)	4.4 (0.5)†	5.1 (0.7)	4.4 (0.6)†	5.0 (0.7)	4.7 (0.5)
PcvCO ₂ (kPa)	6.2 (0.6)	6.2 (0.5)	6.4 (0.5)	5.9 (0.4)†	6.2 (0.5)	6.1 (0.5)	6.5 (0.7)	6.5 (0.6)
S _a O ₂ (%)	99.8 (0.1)	99.8 (0.1)	99.3 (1.7)	99.7 (0.3)	99.3 (1.2)	99.7 (0.6)	99.8 (0.1)	99.8 (0.1)
ScvO ₂ (%)	84.2 (6.6)	80.9 (9.2)	77.0 (8.5)*	71.5 (10.0)*†	76.5 (9.2)*	70.5 (8.3)*†	82.7 (5.7)	79.1 (10.2)
PAO ₂ (kPa)	89.0 (0.8)	89.2 (0.6)	88.7 (1.0)	89.5 (0.6)†	88.5 (0.8)	89.5 (0.7)†	89.0 (1.8)	89.2 (0.7)
A-aO ₂ (kPa)	36.1 (10.4)	35.5 (9.6)	60.8 (9.0)*	62.0 (9.3)*	59.4 (10.6)*	58.3 (9.9)*	38.6 (10.6)	39.0 (12.3)
Qs/Qt (%)	20.7 (4.9)	18.6 (7.2)	27.5 (7.5)*	23.8 (4.4)*†	26.1 (6.9)*	22.1 (5.1)†	20.6 (6.6)	18.9 (7.0)
Pa-E'CO ₂ (kPa)	0.7 (0.4)	0.6 (0.4)	1.0 (0.6)	0.5 (0.3)†	1.2 (0.4)*	0.4 (0.3)†	1.1 (0.5)	0.7 (0.4)†
Estimated Vd/Vt (%)	15.5 (7.9)	14.5 (7.8)	21.8 (10.8)	12.9 (6.6)†	24.6 (8.1)*	11.1 (7.6)†	23.4 (10.8)	17.9 (8.8)*

Values are mean (standard deviation). P_aO₂=arterial oxygen tension, PcvO₂=central venous oxygen tension, P_aCO₂=arterial carbon dioxide tension, PcvCO₂=central venous carbon dioxide tension, S_aO₂=arterial oxygen saturation, ScvO₂=central venous oxygen saturation, PAO₂=alveolar oxygen tension, A-aO₂, alveolar-arterial oxygen gradient, Qs/Qt=intrapulmonary shunt fraction (assuming that central venous oxygen saturation is equal to mixed venous oxygen saturation), Pa-E'CO₂=arterial to end-tidal carbon dioxide tension difference, Vd/Vt=physiological dead space (estimated according to the Hardman and Aitkenhead equation¹⁴), TLV_{baseline}=15 min after TLV in lateral decubitus position, OLV₃₀=after 30 min of OLV; OLV₆₀=after 60 min of OLV, TLV_{end}=15 min after TLV was re-initiated. * *P* < 0.05 vs TLV_{baseline} in each group, † *P* < 0.05 vs group 1:2.

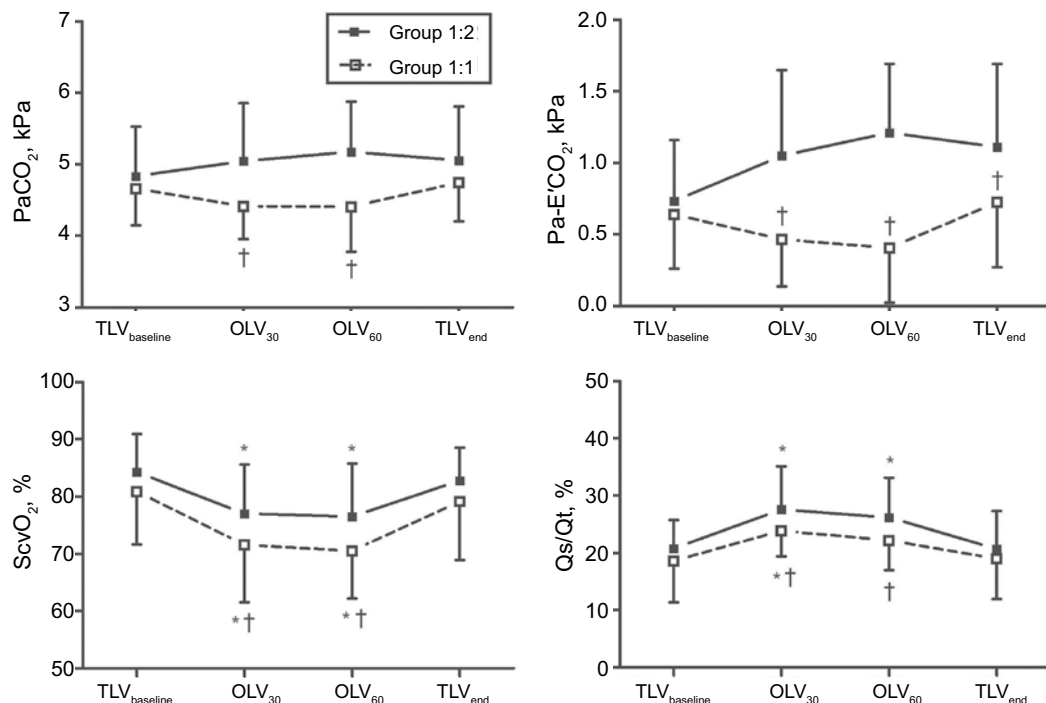


FIGURE 1: Changes in P_aCO₂, Pa-E'CO₂ difference, ScvO₂ and Qs/Qt. P_aCO₂=arterial carbon dioxide tension, ScvO₂=central venous oxygen saturation, Pa-E'CO₂=arterial to end-tidal carbon dioxide tension difference, Qs/Qt=intrapulmonary shunt fraction (assuming that central venous oxygen saturation is equal to mixed venous oxygen saturation). TLV_{baseline}=15 min after TLV in lateral decubitus position, OLV₃₀=after 30 min of OLV, OLV₆₀=after 60 min of OLV, TLV_{end}=15 min after TLV was re-initiated. * *P* < 0.05 vs TLV_{baseline} in each group. † *P* < 0.05 vs group 1:2.

TABLE 3
Respiratory mechanics and haemodynamic data

	TLV _{baseline}		OLV ₃₀		OLV ₆₀		TLV _{end}	
	Group 1:2	Group 1:1	Group 1:2	Group 1:1	Group 1:2	Group 1:1	Group 1:2	Group 1:1
RR, breaths/min	10.9 (1.4)	11.3 (1.1)	10.3 (1.6)	10.4 (1.1)	10.3 (1.4)	10.5 (1.3)	10.4 (1.4)	10.4 (1.2)
P _{peak} , cmH ₂ O	14.4 (2.4)	13.8 (2.4)	23.5 (4.2)*	18.5 (3.2)*†	22.7 (4.3)*	18.6 (3.3)*†	17.0 (5.1)	15.2 (4.4)
P _{plat} , cmH ₂ O	12.6 (3.0)	11.3 (2.2)	19.3 (5.2)*	15.9 (2.9)*†	18.8 (5.2)*	15.8 (3.2)*†	14.9 (5.8)	12.1 (3.9)
P _{mean} , cmH ₂ O	4.2 (1.5)	4.7 (1.0)	6.3 (1.5)*	8.3 (1.7)*†	6.1 (1.7)*	8.1 (1.7)*†	4.5 (1.8)	5.3 (1.6)
C _{dyn} , ml/cmH ₂ O	44.7 (9.5)	48.6 (10.7)	29.8 (11.8)*	36.7 (7.8)*†	30.7 (11.8)*	38.4 (7.9)*†	39.8 (18.9)	43.7 (16.8)
HR, min	68.5 (9.1)	70.7 (11.9)	72.8 (11.7)	74.0 (9.5)	74.1 (10.1)	70.0 (7.1)	74.4 (11.4)	72.6 (10.4)
MAP, mmHg	83.3 (10.8)	88.0 (11.6)	81.4 (10.6)	85.7 (13.0)	82.6 (13.5)	89.1 (10.2)	83.0 (11.9)	85.6 (9.9)
CVP, mmHg	7.9 (3.2)	8.7 (3.4)	9.4 (3.4)	10.7 (3.5)	8.8 (2.7)	9.8 (2.8)	8.0 (3.4)	9.46 (3.0)

RR=respiratory rate, P_{peak}=peak airway pressure, P_{plat}=plateau airway pressure, P_{mean}=mean airway pressure, C_{dyn}=dynamic compliance, HR=heart rate, MAP=mean arterial pressure, CVP=central venous pressure, TLV_{baseline}=15 min after TLV in the lateral decubitus position, OLV₃₀=after 30 min of OLV, OLV₆₀=after 60 min of OLV, TLV_{end}=15 min after TLV was re-initiated. Values are mean (standard deviation). * $P < 0.05$ vs TLV_{baseline} in each group, † $P < 0.05$ vs group 1:2.

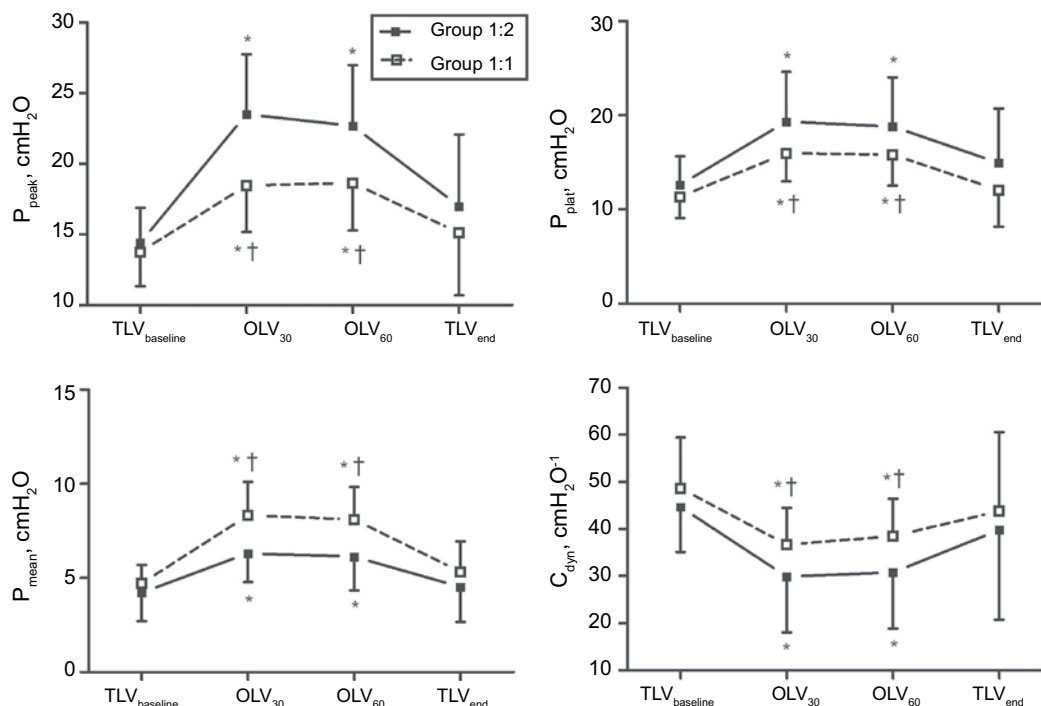


FIGURE 2: Changes in airway pressures and dynamic compliance. P_{peak}=peak airway pressure, P_{plat}=plateau airway pressure, P_{mean}=mean airway pressure, C_{dyn}=dynamic compliance, TLV_{baseline}=15 min after TLV in lateral decubitus position, OLV₃₀=after 30 min of OLV, OLV₆₀=after 60 min of OLV, TLV_{end}=15 min after TLV was re-initiated. * $P < 0.05$ vs TLV_{baseline} in each group. † $P < 0.05$ vs group 1:2.

When the double-lumen tube was well-positioned, OLV resulted in an approximately 55% increase in P_{peak} and a 41% increase in P_{plat} compared with TLV_{baseline}. The increase in P_{peak} may be a sign of conditions associated with endotracheal tube resistance, decreased compliance and increased flow resistance^{15,16}. Although P_{peak} does not reflect peak alveolar pressure^{16,17}, clinically high P_{peak} may

contribute to hyperinflation injury of the ventilated lung during OLV². P_{plat} reflects small airways and alveolar pressure, and there is a significant correlation between P_{plat} and mechanical ventilation-induced barotrauma¹⁸. Our study showed that P_{peak} and P_{plat} were reduced 21 and 16%, respectively, for VCV at an I:E ratio of 1:1 compared with an I:E ratio of 1:2 during OLV; this result is similar to or greater than

the reduction in P_{peak} and P_{plat} reported in previous studies comparing pressure-controlled ventilation (PCV) and VCV at the same tidal volume during OLV^{17,19}. Although the decelerating inspiratory flow method used during PCV is associated with reduction of P_{peak} , PCV has limitations of variability of delivered tidal volume during OLV²⁰. VCV at an I:E ratio of 1:1 with constant inspiratory flow not only reduced P_{peak} and P_{plat} by a similar extent to PCV, but also guaranteed a consistent tidal volume. In this study, P_{mean} was approximately 30% higher for the I:E ratio of 1:1 than for the I:E ratio of 1:2. Low P_{mean} as an index of mean alveolar pressure usually requires the application of extrinsic PEEP for adequate alveolar distension during OLV with low tidal volume²¹. Our study showed that minimal prolonged I:E ratio ventilation effectively increased P_{mean} during OLV. The compliance of the dependent lung is decreased by a reduction in chest wall compliance, lung compression due to gravitational effects, surgical stimuli and high airway pressure in the lateral decubitus position during OLV, and for these reasons atelectasis and alveolar collapse may readily occur in a dependent lung⁴. VCV with an I:E ratio of 1:1 effectively improved dynamic lung compliance during OLV, suggesting that atelectasis and alveolar collapse in the dependent lung might be reduced.

Despite the beneficial effects of VCV with an I:E ratio of 1:1 on respiratory mechanics during OLV compared with VCV with an I:E ratio of 1:2, there was no difference in oxygenation between the two groups. Oxygenation during OLV is affected by the interaction between intrapulmonary shunt, cardiac output, oxygen extraction, venous saturation and haemoglobin level^{4,5}. Lower oxygen content in the venous blood directly results in a decrease in intrapulmonary shunting in VCV with an I:E ratio of 1:1 relative to VCV with an I:E ratio of 1:2⁵. The low intrapulmonary shunt fraction in group 1:1 may be explained by the low venous oxygen tension that contributed to the favourable hypoxic pulmonary vasoconstriction in the non-dependent lung and improvement in V/Q mismatching in the dependent lung⁶. Although we did not directly measure cardiac output in this study, an increase in P_{mean} is known to be the main cause of decreased cardiac output in prolonged I:E ratio ventilation^{9,10}. A decreased cardiac output may lead to a decrease in both intrapulmonary shunt and venous oxygen saturation with a decrease in P_aO_2 during OLV^{5,22}. Central and mixed venous oxygen saturation provides valuable information about the balance between oxygen delivery and oxygen consumption and thus the

adequacy of tissue oxygenation^{23,24}. The lower $ScvO_2$ in VCV at an I:E ratio of 1:1 compared with an I:E ratio of 1:2 may reflect a greater increase in tissue oxygen extraction and consequently more inadequate tissue oxygenation. In our study, there was no difference in preoperative and intraoperative haemoglobin levels between the two groups at each assessment point. For all these reasons, although 1:1 I:E ratio ventilation showed favourable effects on intrapulmonary shunt, decreased cardiac output due to increased P_{mean} and inadequate tissue oxygenation may have resulted in the lack of a significant difference in P_aO_2 between the two groups.

OLV in the lateral decubitus position generally increases dead space and $Pa-E'CO_2$ difference, and requires a greater increase in minute ventilation to maintain the same P_aCO_2 ²⁵. However, as has been observed in previous studies of inverse ratio ventilation^{9,10,12}, VCV with an I:E ratio of 1:1 improved the efficiency of alveolar ventilation during OLV compared with VCV with an I:E ratio of 1:2 in our study, resulting in a lower $Pa-E'CO_2$ difference and a reduced physiologic dead space. These results indirectly reflect a reduction in the amount of atelectasis and an improvement in V/Q mismatching in the dependent lung^{26,27}. Interestingly, most previous studies of inverse ratio ventilation used a more prolonged I:E ratio than 1:1 and PCV with decelerating inspiratory flow in the supine position^{10,11,28}, but our results showed that VCV with only a minimal prolonged I:E ratio could result in improved CO_2 elimination and lower ventilatory requirements during OLV.

This study has several limitations. First, this study was conducted in patients with normal preoperative pulmonary function and none of the patients included in the study had any significant airflow obstruction preoperatively that would increase the risk of air trapping during OLV. Also, patients were ventilated with 100% oxygen without the application of extrinsic PEEP. Second, we calculated the intrapulmonary shunt fraction based on $ScvO_2$, not SvO_2 , and did not measure cardiac output since our patients with normal cardiac function did not require a pulmonary artery catheter. Third, we could not measure the amount of intrinsic PEEP by airway occlusion technique during surgery²⁹. We used a relatively large tidal volume (8 ml/kg) during OLV because this study was conducted without extrinsic PEEP. Furthermore, we did not measure endobronchial or systemic cytokines (inflammatory mediators) which have been known to be associated with lung injury³⁰. Therefore, we cannot be sure whether the associated increase in P_{mean} during VCV with an I:E ratio of 1:1

would have induced more lung injury than an I:E ratio of 1:2. However, there were no patients with respiratory failure requiring ventilator care, air leak with pneumothorax or prolonged chest tube drainage in postoperative periods who underwent surgery using an I:E ratio of 1:1.

In conclusion, it should be emphasised that VCV with an I:E ratio of 1:1 is a viable alternative ventilatory strategy for OLV with regard to the effect of reducing P_{peak} and P_{plat} , improvement in dynamic compliance and an increase in efficiency of alveolar ventilation without significant haemodynamic changes during OLV in the lateral decubitus position, although this did not result in substantial improvement in oxygenation.

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